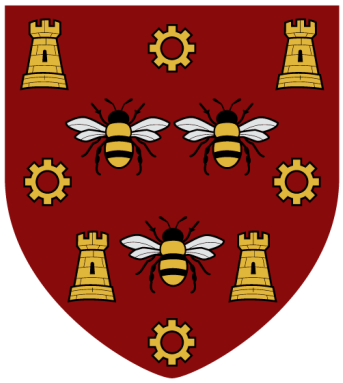
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Dialogue Through Play: Games as a Bridge for Citizen Engagement

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ABSTRACT

Background: Digital games are increasingly used as tools for civic and societal engagement, particularly to reach younger and digitally literate publics on complex issues such as climate change. While such approaches can facilitate large-scale participation and raise awareness, their capacity to support deliberative forms of civic engagement within real-world policy contexts remains insufficiently examined.

Objectives: This study examines whether and how survey-based engagement embedded within a commercial digital game can move beyond surface-level participation towards forms of engagement relevant to policy deliberation.

Methods: The paper reports on an exploratory case study (GCS1) conducted within the GREAT project, in which a multiple-choice climate policy survey was embedded into a commercial multiplayer game. The design prioritised scale and accessibility over deliberative depth, enabling large-scale, anonymous data collection from a predominantly young and digitally literate population. Descriptive and inferential analyses were used to examine patterns of engagement and policy preferences across demographic groups.

Results and Conclusions: The embedded survey achieved high levels of participation, demonstrating the feasibility of game-based data collection at scale and the potential of games to introduce climate policy issues to digitally engaged publics. However, expressed engagement and policy preferences did not provide evidence of deliberative reasoning, reflecting constraints associated with survey design, platform demographics and the absence of reflective or dialogic mechanisms. Game-based surveys are therefore best understood as complementary entry points for civic engagement rather than standalone tools for democratic deliberation, highlighting the need to integrate games within broader deliberative and institutional frameworks.

1 | Introduction

There is a growing interest in deploying games as instruments for improving public engagement in policymaking, including climate policy. However, engagement in this context broadly encompasses both strategic and reflective dimensions. Strategic engagement refers to participation motivated by external incentives, whereas reflective engagement involves deeper cognitive and affective processing, leading to shifts in understanding or attitudes. Games can foster both, but their capacity to support deliberative engagement, understood here as processes through which participants reflect on policy trade-offs, consider alternative perspectives, and articulate informed preferences, remains insufficiently examined in large-scale, real-world settings. This paper examines survey-based civic engagement embedded in commercial digital games, with a focus on the conditions under which such participation supports reflection relevant to policy discourse.

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Lay Description

- What is currently known about this topic?
  - Digital games are increasingly used to engage citizens with climate and policy issues.
- What does this paper add?
  - This study shows that surveys embedded in a commercial game can reach large audiences but have limited capacity to support reflective policy deliberation.
- Implications for practice/or policy:
  - Game-based surveys work best as entry points for engagement when linked to wider deliberative and policy processes.

Two central research questions underpin this paper: How effective are games as a communication bridge for citizen engagement in policymaking? What are the limitations and challenges of gathering data through games-based methodologies? Addressing these questions, the paper presents an empirical case study used to interrogate the conditions under which game-based engagement may enable, constrain or fall short of deliberative participation. Drawing on data gathered within the GREAT Project UNDP case study (GCS1), the paper evaluates the strengths and limitations of using games as environments for civic data collection and policy communication, with a particular focus on climate-related issues. Persuasive game theory suggests that games can influence attitudes and behaviours through interactive storytelling and decision-making mechanics (Bogost 2010; Kolek et al. 2023); however, evidence regarding their role in sustained civic reasoning and integration into institutional decision-making remains mixed and contested. While games have been shown to spark dialogue on social issues, their ability to move beyond engagement towards deliberative contribution within policy processes remains underexplored. This gap is particularly salient given the increasing reliance on digital participation mechanisms to reach publics that are otherwise disengaged from formal democratic processes.

The GREAT Project explores innovative approaches to engaging citizens in policy development, particularly in response to the climate emergency. As part of this initiative, GCS1 focuses on the use of games-based methodologies as a communication tool to elicit public feedback. By embedding surveys within widely played commercial games, the study expanded its reach to a young and digitally literate audience that is typically underrepresented in traditional policy consultations. This design prioritised scale and accessibility over depth of interaction, creating an opportunity to examine the trade-offs inherent in large-scale, game-mediated engagement. While this approach demonstrated the potential of gaming environments to introduce policy issues and stimulate participation, its capacity to support deliberative engagement, where individuals critically evaluate issues and reflect on competing policy options, remains uncertain and is a central concern of this study.

Youth face growing environmental challenges yet remain marginalised in climate policy discussions (Wray-Lake et al. 2010; Perlaviciute and Squintani 2020; Arnot, Thomas, Pitt, and Warner 2024). Despite the importance of youth civic engagement, systemic exclusion (Ruppel and Houston 2023; Arnot, Thomas, Pitt, McCarthy, and Warner 2024) and declining participation in traditional political processes (Dahlgren 2013; Barrett 2018) persist, despite climate policy decisions disproportionately affecting younger generations (Corner et al. 2015; Baldwin et al. 2023). At the same time, moral disengagement driven by the scale and complexity of the climate crisis can inhibit sustained action (Bandura and Cherry 2020; Stoll-Kleemann and O'Riordan 2020). While youth activism has received increasing scholarly attention (Neas et al. 2022), broader patterns of everyday engagement and participation remain underexplored. Research on youth climate engagement has largely focused on communication strategies (Barnosky et al. 2016; Amelung et al. 2016; Roosen et al. 2018; Sippel et al. 2022), while the mechanisms through which young people form, express and refine policy-relevant preferences are less well understood.

Digital media, including games, offer interactive ways to connect youth with societal issues (Bleumers et al. 2012; Stewart et al. 2013) and foster civic participation (Gordon and Baldwin-Philippi 2014). Civic games and platforms such as Community PlanIt, alongside initiatives in Kenya and Nigeria, demonstrate the potential of game-like systems to support participation (Fisher 2020), although scaling such impact remains challenging (Kahne et al. 2009; Stewart et al. 2013). These challenges raise questions not only about reach, but about whose voices are amplified, whose are excluded, and how participation is shaped by platform design.

Games also enable researchers to capture player behaviours and decision-making in naturalistic digital environments (Gundry and Deterding 2018). The GREAT project builds on this tradition, alongside initiatives such as Sea Hero Quest, iCivics and Foldit, which have demonstrated the value of games for scientific and civic research (Hyde et al. 2016; Spiers et al. 2017; Blevins et al. 2014; Bauer et al. 2017). Political and ethics-based games, including Democracy 3 and This War of Mine, have shown how gameplay can surface engagement with social issues (Dahya 2009; Neys et al. 2012; Dunwell 2012). However, these approaches frequently overrepresent young, digitally literate male players while marginalising others (Ducheneaut 2010; Flanagan and Kaufman 2016). Such structural biases are not incidental but shape both participation and the data generated, complicating claims about deliberation and representativeness (Gundry and Deterding 2018; Leite et al. 2019; Øygardslia and Ask 2022).

Integrating game-based features into civic engagement strategies aligns with the digital practices of younger generations (Nelson and Lei 2018; García-Galera et al. 2018). Traditional participatory mechanisms, such as town hall meetings and participatory budgeting, have increasingly incorporated digital platforms to broaden access and participation (Vromen and Collin 2010; de Oliveira 2017). Platforms such as CrowdLaw and Decidim illustrate how digital tools can support large-scale participation in policy discussion and decision-making (Thiel et al. 2019; Feddersen and Santana 2022; Setty and Dobson 2023). However, digital participation does not inherently guarantee equitable or deliberative engagement (Çiçek 2024). Within this landscape, GCS1 employs a survey-based methodology embedded in

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commercial games to examine what forms of engagement are made possible and what forms are constrained by such environments. While games offer scalability and accessibility, concerns persist regarding the validity and reliability of in-game data (Gundry and Deterding 2019), as well as the influence of game mechanics on responses and the generalisability of findings to policy contexts (Gundry and Deterding 2022).

Games have increasingly become platforms for civic engagement, offering interactive spaces to explore social and environmental issues. Serious and persuasive games encourage reflection on moral, ethical, and civic dilemmas (Lee et al. 2013), and titles such as Democracy 3, SimCity, This War of Mine, and Papers, Please have been used to foster awareness and empathy (Peng et al. 2010; Shliakhovchuk 2024). However, these narratives often simplify complex policy realities (Bereitschaft 2016), and their impact on real-world civic reasoning and action remains difficult to assess empirically (Ryan et al. 2020). By examining a large-scale, survey-based game intervention, this study contributes an empirically grounded assessment of the deliberative affordances and limitations of games as tools for civic engagement within climate policy discourse.

2 | Materials and Methods

The GREAT project's case study programme explored digital interventions in public engagement, participatory research, and policy discourse. Although the wider programme adopted a multiple-case study design, this paper reports on a single exploratory case (GCS1), used analytically to examine the affordances and constraints of game-based survey methodologies for civic engagement. Case study approaches are well suited to examining complex, context-dependent digital interventions where experimental control is limited (Zainal 2007; Johnson 2006). The study offers analytical insight into how embedding research within commercial gaming environments shapes patterns of engagement.

This research assessed the effectiveness of digital tools while critically addressing ethical, practical, and methodological challenges associated with embedding surveys within entertainment contexts. By integrating research into an existing digital ecosystem, the study examined whether such approaches could lower participation barriers and generate scalable insights into public attitudes on climate policy. GCS1 was explicitly exploratory in nature and followed an eight-step research framework designed to ensure procedural coherence while allowing methodological flexibility appropriate to an emergent, real-world intervention. The framework encompassed planning, design, data collection, analysis, and evaluation, and was developed collaboratively with sponsors, academic leads and stakeholders. Iterative refinement enabled the research team to respond to emerging challenges while meeting ethical, consent and data management requirements.

Central to the study design was the decision to prioritise reach and accessibility over depth of interaction. Traditional public consultation methods often rely on self-selecting participants and static surveys, which can limit engagement among younger demographics. To address this, GCS1 embedded surveys within the commercial multiplayer game *SMITE*, leveraging its large, globally distributed player base and established in-game communication infrastructure to access a digitally literate, youth-dominated audience. This platform choice reflects a conscious decision to work within an existing entertainment ecosystem, while acknowledging trade-offs in deliberative depth.

Survey prompts were aligned with player interaction patterns to minimise disruption to gameplay while capturing contemporaneous responses. Participation was voluntary, with players directed to an external survey via QR codes (Figure 1). The survey consisted of 13 multiple-choice questions addressing climate policy domains including energy, transportation, biodiversity, food systems, economic measures and disaster preparedness, alongside demographic questions (age, gender, education). No open-ended questions or follow-up interviews were included, reflecting a deliberate methodological trade-off in favour of standardisation, comparability and scalability. Questions were phrased using simple, accessible language to minimise comprehension barriers. Anonymous data was collected, enabling large-scale aggregation across geographic regions.

This methodological design imposes constraints that influence the insights the study produces. As participation was voluntary, the resulting sample cannot be assumed to represent either the full gaming population or the general public. Rather than treating representativeness as an objective, the study focuses on understanding engagement patterns within a specific digitally active cohort. In addition, the exclusive use of multiple-choice questions limited insight into participants' underlying reasoning processes, allowing the case study to assess expressed preferences and issue salience, but not deliberative reasoning or opinion formation. The study uses multiple-choice survey instruments to collect large-scale data within an entertainment context. While this design supports comparability and scalability, it may encourage familiar or socially desirable responses (Andersen and Mayerl 2019; Cerri et al. 2019; King 2022). Attention in gameplay is distributed across entertainment and survey tasks, which shapes the cognitive demands of participation. The study does not include a control group; its focus is on identifying the conditions under which policy-relevant engagement can be elicited at scale rather than on measuring causal effects.

These limitations are not incidental but integral to the methodological choices made. Future research could address them through mixed-method designs incorporating open-ended responses, deliberative prompts, or comparative deployment across gaming and non-gaming platforms. However, within the scope of this study, the method is used to interrogate the boundary between engagement and deliberation, rather than to resolve it. This framing allows the analysis to contribute to methodological debates on digital participation while avoiding overextension of empirical claims.

3 | Results

This case study assessed the effectiveness of a game-based survey methodology in capturing public perspectives on climate policy, with a particular focus on patterns of engagement across

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FIGURE 1 | Placement of the survey prompt within the *SMITE* interface.

demographic groups. Data were analysed using descriptive statistics to examine response distributions across policy domains and demographic variables. Given the exclusive use of multiple-choice questions, the analysis focuses on expressed preferences and issue salience. Statistical tests were used to identify significant associations between demographic variables and policy responses.

A total of 2365 participants responded to at least one survey question, representing a 93.8% engagement rate. This high level of participation indicates that embedding surveys within a commercial game environment can successfully elicit responses at scale. However, participation and response frequency should be interpreted as indicators of engagement and preference expression, rather than evidence of deliberative policy reasoning.

Table 1 summarises responses to core climate policy questions, including perceived urgency of climate action and support for specific intervention domains. Overall, 56.2% of participants supported urgent climate action (Table 1, Q1), while smaller proportions favoured gradual change or no action. This distribution suggests broad receptiveness to climate action among respondents. Similarly, 70.1% of respondents agreed that games are effective for raising climate awareness (Table 2, Q9), reinforcing the communicative potential of game-based approaches.

3.1 | Demographic Patterns in Policy Preferences

Age was significantly associated with support for several climate policy domains. Younger respondents (18–35) were more likely to support urgent interventions across energy, transport, and urban planning, while older respondents (60+) showed significantly higher resistance, particularly towards clean electric and clean freight transport, and ocean and waterway protection (Table 1, Q2–Q4). For example, resistance to clean energy transport among the 60+ group deviated strongly from expected values (Adjusted Residual = 9.7). These patterns indicate generational differences in expressed policy preferences within the sample. Chi-square analyses confirmed significant associations between age and support for clean electric and freight transport options ($\chi^2(6) = 102.75$, $p < 0.001$) and urban/rural planning ($\chi^2(3) = 28.85$, $p < 0.001$). While under-18 respondents showed relatively high support for clean transport (64.9%), their lower expressed urgency in other domains suggests heterogeneity even within younger cohorts, illustrating the limits of treating youth engagement as uniform.

Education level was also significantly associated with climate policy preferences. Respondents with higher levels of education were consistently more supportive of urgent climate action and sustainability measures, while those with minimal or no formal education showed higher support for inaction (Adjusted Residual = 9.9). Education therefore emerged as a consistent correlate of expressed climate concern across multiple policy domains.

3.2 | Gender and Engagement With Climate Action

Gender differences were evident across several policy domains. While support for renewable energy was high across all genders and not statistically significant, women were significantly more likely than men to support improved urban and rural planning ($\chi^2(2) = 6.579$, $p = 0.037$) and clean

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TABLE 1 | Climate action preferences (Q1–Q7).

Question	Most supported action	Second most supported action	Third most supported action
Q1: What should the world do about climate change?	56.2% Do everything necessary, urgently.	35.3% Act slowly whilst we learn.	8.9% Do nothing.
Q2: To address the climate crisis, how should your country improve transport?	63.7% Improve urban/rural planning.	59.5% Clean electric transport.	47.9% Clean freight transport.
Q3: To address the climate crisis, what should your country do about energy?	75.8% Use renewable energy.	53.9% Stop burning polluting fuels.	47.8% Reduce energy waste.
Q4: To address the climate crisis, what do you think your country should do about nature?	83.4% Keep oceans/waterways healthy.	78.5% Conserve forests/land	49.5% Support local communities.
Q5: To address the climate crisis, what should Governments do about farms and food?	74.3% Reduce food waste.	73.9% Climate-friendly farming.	31.3% Promote plant-based diets.
Q6: To address the climate crisis, what should Governments do about the economy?	69.6% Make companies pay for pollution.	65.9% Invest in green business/jobs.	58.2% Require transparency on products.
Q7: How can your country better protect people from extreme storms, flooding, droughts, forest fires, and other climate impacts?	74.2% Build infrastructure and conserve nature.	66.3% Early warning systems.	57.0% Affordable insurance.

TABLE 2 | Perceptions on games and climate change (Q8–Q10).

Question	Most supported response	Percentage
Q8: Do you think games can contribute to resolving climate change?	Yes	57.5%
Q9: In your view, how do you think games could best help tackle climate change?	Raise awareness	70.1%
Q10: What climate topics do you think games can best cover?	All of the above	43.0%

transport transitions ($\chi^2(2) = 12.263$, $p = 0.002$). Women also showed higher support for urgent climate action overall ($\chi^2(4) = 43.841$, $p < 0.001$). These differences reflect gendered patterns in policy preferences rather than variation in deliberative engagement. Non-binary respondents displayed distinct response patterns, including a higher-than-expected preference for no action (Adjusted Residual = 5.0). Given the small size of this subgroup, these findings should be interpreted as indicative rather than representative.

TABLE 3 | Demographics (Q11–Q13).

Question	Most common group	Percentage
Q11: Age	18–35	84.9%
Q12: Gender	Male	85.2%
Q13: How old were you when you left education?	20 or over	37.9%

3.3 | Perceptions of Games as Civic Tools

Respondents generally expressed positive views about the role of games in climate engagement. Female respondents were more likely than males to believe games could raise awareness, support education, and enable expression of climate-related views (Table 2, Q8–Q10). These findings point to the perceived legitimacy of games as communicative and engagement-oriented tools.

3.4 | Geographic and Demographic Representation

Table 3 presents the demographic characteristics of respondents, with the majority aged 18–35 and a predominance of male participants, reflecting known gaming demographics

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(Carcelén-García et al. 2023). Table 4 demonstrates that most responses originated from North America and parts of Europe, particularly the United States. This concentration reflects platform demographics (Meriläinen 2023; Drenten et al. 2023), rather than targeted sampling and limits the generalisability of the findings to global climate publics. The underrepresentation of older adults, women and participants from climate-vulnerable regions highlights structural constraints in using commercial games as inclusive civic engagement platforms.

Across policy domains, demographic variables showed statistically significant associations with expressed preferences, particularly for transport (Table 1, Q2), energy (Table 1, Q3), and nature-related interventions (Table 1, Q4). Age and education were more consistently associated with these responses than gender, as indicated by chi-square tests and adjusted residuals reported for transport, urban/rural planning, and urgency of climate action (χ^2 tests, $p < 0.001$), reinforcing their relevance as structural correlates of climate policy preferences within this sample.

4 | Synthesis

The findings demonstrate that game-based surveys can achieve high levels of participation and capture broad patterns of climate policy preferences among digitally engaged populations. At the same time, expressed support for climate action reflects issue salience and normative alignment rather than deliberative engagement. Demographic and platform biases shape both who participates and which policy priorities are foregrounded. Importantly, identifying where deliberative engagement does not emerge under conditions of scale is itself analytically valuable, as it clarifies the design and institutional requirements necessary for deliberation to occur in digital participation systems.

5 | Discussion

This case study illustrates both the potential and the limits of game-based survey methodologies as a mechanism for citizen engagement in policymaking. While high participation levels demonstrate that digital games can facilitate large-scale data collection, the findings highlight a persistent tension between engagement at scale and the conditions required for deliberative participation. The central issue concerns how games enable meaningful civic engagement within the constraints of commercial gaming environments.

Digital platforms have been shown to enhance accessibility and participation (Çiçek 2024), yet the distinction between engagement and deliberation remains critical. Prior research suggests digital tools can prompt discussion and awareness (Gonzalez-Mohino et al. 2023; Sleigh et al. 2024), while often falling short of supporting sustained, reflective policy reasoning (Davies and Procter 2020; Çiçek 2024). In GCS1, embedding a policy-focused survey within gameplay increased exposure to climate issues and elicited clear expressions of policy preference, capturing patterns of opinion and issue salience within the player population.

The distinction between strategic engagement and reflective engagement is central to interpreting the results. Responses reported in Table 1 indicate strong support for climate interventions across several domains, aligning with broader public sentiment (Vuong et al. 2024). These patterns demonstrate issue salience and normative alignment, but do not constitute evidence of deliberation, which would require participants to engage with trade-offs, uncertainty, and competing values. The absence of embedded deliberative mechanisms, such as iterative dialogue, justification of choices, or exposure to counterarguments, limits the extent to which the observed engagement can be interpreted as deliberative.

Multiple-choice responses shape the type of engagement observed in GCS1. The format captures patterns of policy preference and issue salience across the player population, although it may amplify familiar or socially desirable responses given the entertainment context (Andersen and Mayerl 2019; Cerri et al. 2019; King 2022). These findings reflect expressed opinions and priorities rather than evidence of reflective policy reasoning or attitude change.

Demographic patterns observed in the data further illustrate the structural constraints of game-based engagement. The predominance of younger, male participants reflects the demographics of the *SMITE* player base rather than the broader public (United Nations 2024). This concentration shapes both whose voices are captured and which policy priorities are foregrounded, raising important questions about representativeness and deliberative legitimacy (Hartnett 2022; Reddick et al. 2024; Gao et al. 2017; Cote 2017). While younger participants showed strong support for climate action, older adults, who often exert greater influence in formal policymaking contexts, were underrepresented (El Khoury et al. 2023; Čapienė et al. 2024). These asymmetries highlight the risk that game-based engagement may amplify already mobilised or digitally active groups while marginalising others.

At the institutional level, the impact of game-based engagement remains constrained by prevailing norms in policymaking. Despite growing interest in digital participation, policymakers continue to privilege established consultation methods such as surveys, expert panels, and focus groups (Wibeck and Neset 2020). As a result, data generated through

TABLE 4 | Geographic distribution of respondents.

Country	Sessions
United States	2119
Canada	289
Brazil	225
United Kingdom	184
Mexico	157
Spain	156
Argentina	138
Germany	126
France	114
Russia	93
Colombia	50

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games risks being perceived as supplementary or informal, limiting its integration into decision-making processes (Yau et al. 2024). Although civic gaming theories emphasise its potential for political engagement, particularly among younger demographics, integrating gaming data into formal policymaking remains challenging (Schrier 2018; Kleiman and Janssen 2018).

Beyond identifying these constraints, the findings point to several feasible pathways through which game-based engagement could inform policymaking without replacing established deliberative mechanisms. Game-based surveys may function as agenda-setting tools, surfacing issue salience among digitally engaged publics; as exploratory inputs during early policy scoping phases; or as supplementary evidence alongside formal consultations. In such roles, game-based data would not serve as deliberative outcomes, but as signals that inform where deeper, structured deliberation may be most productively directed.

Participants' perceptions of games as tools for climate engagement (Table 2) suggest games are widely viewed as effective for awareness-raising and education. However, perceived effectiveness does not equate to civic impact. The absence of a clear link between in-game engagement and real-world participation illustrates a missing connective mechanism between digital interaction and policy influence. Without integration into broader deliberative systems, insights generated through gameplay risk remaining episodic rather than cumulative.

These findings suggest game-based surveys are well suited to initiating climate dialogue and capturing broad patterns of opinion but are insufficient on their own to support deliberative policymaking. Bridging this gap would require design approaches that embed reflection and justification within gameplay, as well as institutional arrangements that connect digital engagement to formal deliberative processes. The value of game-based engagement therefore lies not in replacing existing methods, but in complementing them within a broader participatory ecosystem. Empirically, this study contributes large-scale evidence on preference expression in game-mediated engagement; conceptually, it clarifies the boundary between participation and deliberation in digital civic tools.

6 | Conclusion

This case study demonstrates that digital games can function as accessible entry points for large-scale, low-threshold citizen engagement with climate policy, enabling large-scale participation without reliance on traditional recruitment mechanisms. By embedding surveys within an existing multiplayer game, GCS1 reached a predominantly young and digitally literate audience, illustrating the feasibility of game-based data collection at scale. At the same time, the findings illustrate the limits of such approaches when evaluated through a deliberative lens.

While games can facilitate engagement and raise awareness, this study shows that engagement should not be conflated with deliberation. Demographic biases, the predominance of multiple-choice responses, and the absence of mechanisms for reflection or dialogue constrain the extent to which game-based surveys can support deliberative reasoning or informed policy contribution. As such, game-based engagement is best understood as a complementary mechanism, useful for surfacing issue salience and broad preferences, but insufficient as a standalone tool for democratic deliberation.

A key implication of this study is the need to move beyond large-scale data capture towards models that integrate games within broader deliberative frameworks. Hybrid approaches, where games introduce policy issues and elicit initial responses, followed by structured deliberation in complementary forums, offer a more promising pathway for meaningful civic engagement. Such models could mitigate the risk of superficial participation while preserving the accessibility and reach that make games attractive engagement tools.

The demographic composition of participants highlights enduring questions about inclusion and representation. Future research should systematically examine whether observed biases are intrinsic to game genres or reflect design and recruitment choices. Comparative studies across platforms, genres, and engagement mechanisms would help clarify how different configurations shape participation and deliberative potential. In addition, experimental designs comparing traditional surveys, embedded game interactions and deliberative digital forums could provide clearer evidence of how design choices influence civic reasoning.

Beyond methodological refinement, there remains a pressing challenge of institutional integration. For game-based engagement to contribute meaningfully to policymaking, explicit pathways must exist to connect digital insights with decision-making processes. This may involve partnerships between researchers, game developers and policymakers or the development of intermediary structures that translate engagement data into deliberative inputs. Without such mechanisms, game-based engagement risks producing large volumes of data with limited policy relevance.

GCS1 illustrates both the promise and the constraints of games as a bridge for citizen engagement. Games can lower barriers to participation and introduce complex policy issues in accessible ways, but their democratic value depends on how they are embedded within wider deliberative ecosystems. Future work should therefore focus not on whether games can engage citizens, but on how game-based engagement can be designed and situated to support reflection, deliberation and sustained civic agency.

Author Contributions

Rebecca Harris: conceptualisation, methodology, data interpretation, writing (original draft), writing (review and editing). **Paul Hollins:** funding acquisition, project administration, supervision. **Anchal Garg:** data management, formal analysis. **Pradeep Hewage:** data management. **Paul Watson:** supervision. **Celestine Iwendi:** project administration. **Jude Ower:** data curation, resources, project administration. **Joost Schuur:** data curation, resources, project administration.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in Zenodo at https://zenodo.org/communities/101094766/records?q=&l=list&p=1&s=10&sort=newest.

Peer Review

For transparency, the peer review documents associated with this article are available at https://doi.org/10.1002/jcal.70236.

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